

1. $m = 0.02 \text{ kg}$, $T_1 = 17 + 273.15 \text{ K}$, $T_2 = 27 + 273.15 \text{ K}$, $M_{\text{He}} = 4 \text{ kg/mol}$

(1) V 不变 (2) p 不变 (3) $Q = 0$

求 ΔU , Q , A

解: (1) $A = \int_1^2 p dV = 0$

$$Q = \int_{T_1}^{T_2} C_{v,m} n dT = C_{v,m} \frac{n}{M} (T_2 - T_1)$$

$$= \frac{3}{2} R \times \frac{n}{M} (T_2 - T_1)$$

$$= \frac{3}{2} \times 8.31 \times \frac{0.02}{4 \times 10^{-3}} \times (27 - 17)$$

$$= 623.25 \text{ J}$$

$$\Delta U = A + Q = 623.25 \text{ J}$$

(2) $A = \int_1^2 p dV = p(V_2 - V_1) = nR(T_2 - T_1)$

$$= -\frac{0.02}{4 \times 10^{-3}} \times 8.31 \times 10 = -415.5 \text{ J}$$

$$\Delta U = \int_{T_1}^{T_2} C_{v,m} n dT = 623.25 \text{ J}$$

$$Q = \Delta U + A = 623.25 + 415.5 = 1038.75 \text{ J}$$

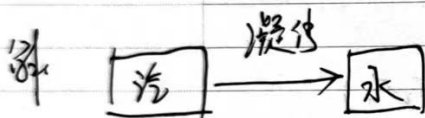
$$(3) \quad Q=0$$

$$\Delta U = \int_1^2 C_{v,m} \nu dT = 623.25 \text{ J}$$

$$A = \Delta U - Q = 623.25 \text{ J}$$

$$14. \quad T = 24 + 273.15 \text{ K} \quad p = 2982.1 \text{ Pa} \quad \text{时} \quad H_{\frac{1}{2}} = 2545.0 \text{ kJ/kg}$$

$$H_{\text{水}} = 100.59 \text{ kJ/kg} \quad \text{求凝结热}$$



$$Q = H_{\text{水}} - H_{\text{汽}} = 100.59 - 2545.0 \\ = -2444.41 \text{ kJ/kg}$$

凝结热: 1kg 汽 释放 2444.41 kJ 热 凝结为水

$$15. \quad p = 1.01 \times 10^5 \text{ Pa} \quad T_1 = 300 \text{ K} \quad T_2 = 1200 \text{ K}$$

$$C_{p,m} = a + bT \quad a = 2.5 \times 10^4 \quad b = 5.92$$

$$\Delta H = (\Delta Q)_p = \int_{T_1}^{T_2} C_{p,m} dT = C_{p,m} (T_2 - T_1) \\ = \int_{T_1}^{T_2} (a + bT) dT$$

$$= a(T_2 - T_1) + \frac{1}{2} b(T_2^2 - T_1^2)$$

$$= (T_2 - T_1) \left[a + \frac{1}{2} b(T_2 + T_1) \right]$$

$$= (1200 - 300) \left[2.3 \times 10^4 + \frac{1}{2} \times 5.92 \times (1200 + 300) \right]$$

$$= 2.47 \times 10^7 \text{ J/mol}$$